

PCME-RAM (Phonon-Coupled Magnetoelectric RAM)

Complete Technical Specification & Research Whitepaper (TRL 3 / TRL 4)

Document ID: OOA-SPEC-2026-PCME-V8

Lead Architect: Emanuel Schaaf

Architecture Paradigm: Open Origin Architecture

Classification: Scientific Research Whitepaper, Concept Architecture, and Preliminary Modeling Framework

1. Executive Summary & Core Directive

Crucial Technical Context:

Several performance metrics in this document (bandwidth, yield, costs, heat flux density, and latency) represent theoretical research goals and modeling assumptions that have not yet been industrially validated. They define the development framework for Technology Readiness Levels (TRL) 3 through 6.

The PCME-RAM concept investigates the functional decoupling of information transmission from electrical charge transfer:

- Vertical Data Transfer:** Partial substitution of signal-carrying copper through-silicon vias (TSVs) with highly directional acoustic wave fields. *Note: Power Delivery Networks (PDN), reference signals, and clock networks will inherently continue to require metallic pathways.*
 - Non-Volatile Memory Cells:** Utilization of magnetoelectric heterostructures that switch purely via strain control, eliminating Joule heating.
 - Integrated Microfluidics:** Direct heat dissipation utilizing microscopic capillary effects within the wafer stack.
 - Enhanced Packaging Tolerances:** Significant increase in alignment tolerances achieved through elastic coupling layers.
-

2. Physical and Mathematical Core Models

2.1 Elastodynamics and Phonon Transit Time

The propagation of acoustic longitudinal waves in monocrystalline silicon is governed by the elastic wave equation:

$$\rho_{Si} \frac{\partial^2 u_i}{\partial t^2} = c_{ijkl} \frac{\partial^2 u_k}{\partial x_j \partial x_l}$$

The phase velocity is given by $v_{Si} = \sqrt{\frac{c_{11}}{\rho_{Si}}} \approx 8433 \text{ m/s}$. The signal transit time through a 16-layer stack with a total thickness of $100 \mu\text{m}$ is calculated as:

$$t_{prop} = \frac{d_{total}}{v_{Si}} \approx 11.86 \text{ ps}$$

2.2 Acoustic Impedance and Transmittance

To validate phonon transmission across layer boundaries, the acoustic impedance $Z = \rho v$ must be matched. The reflection coefficient R and the transmission coefficient T at the interface between silicon (Z_1) and the h-BN epoxy coupling matrix (Z_2) are defined as:

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$$

$$T = 1 - R$$

2.3 Acoustic Attenuation

The dissipation of signal energy across alternating layer boundaries (silicon and polymer) follows an exponential attenuation law. The amplitude decay $A(x)$ over the distance x is:

$$A(x) = A_0 e^{-\alpha x}$$

Similarly, the acoustic power $P(x)$ decays according to:

$$P(x) = P_0 e^{-2\alpha x}$$

Where α represents the frequency-dependent effective attenuation coefficient of the heterogeneous stack. This model numerically dictates the maximum feasible number of memory layers before the signal loss falls below the thermal noise floor of the FDM channel.

2.4 Butterworth-Van-Dyke (BVD) Equivalent Circuit

To simulate the piezoelectric AlN transducer, the BVD equivalent circuit is utilized. It models the resonance via a static capacitance C_0 in parallel with a motional branch (C_m, L_m, R_m). The frequency-dependent impedance $Z(\omega)$ is given by:

$$Z(\omega) = \frac{1}{j\omega C_0 + \frac{1}{R_m + j\omega L_m + 1/(j\omega C_m)}}$$

2.5 Signal Crosstalk

Interference between adjacent acoustic channels is quantified by the crosstalk measure XT in decibels. It compares the induced noise voltage of the victim channel (V_{victim}) with the signal voltage of the aggressor channel ($V_{aggressor}$):

$$XT = 20 \log_{10} \left(\frac{V_{victim}}{V_{aggressor}} \right)$$

2.6 Magnetization Dynamics (LLG Equation)

Switching time, damping, and relaxation of the memory cell are modeled using the Landau-Lifshitz-Gilbert (LLG) equation:

$$\frac{d\mathbf{M}}{dt} = -\gamma(\mathbf{M} \times \mathbf{H}_{eff}) + \frac{\alpha}{M_s} \left(\mathbf{M} \times \frac{d\mathbf{M}}{dt} \right)$$

2.7 Error Propagation and Uncertainty Model

To scientifically evaluate the theoretical projections, the combined standard uncertainty u_c for all derived functions f is determined via:

$$u_c = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} u_i \right)^2}$$

2.8 Acoustic Multiplexing Mitigation: Phononic Metamaterials and Impedance Matching

To overcome the critical high-risk challenges of acoustic attenuation (signal loss) and resonance-induced crosstalk, the PCME-RAM architecture implements two advanced wave-engineering techniques derived from optical physics and metamaterial science.

2.8.1 Quarter-Wave Acoustic Impedance Transformer

To prevent the exponential attenuation caused by reflection at the Silicon-Polymer boundaries ($Z_1 \rightarrow Z_2$), a thin-film matching layer is introduced. Analogous to optical anti-reflective coatings, we utilize a quarter-wave acoustic transformer. The required acoustic impedance Z_{match} of this intermediate layer is calculated as the geometric mean of the two surrounding media:

$$Z_{match} = \sqrt{Z_{Si} Z_{Polymer}}$$

To achieve destructive interference of the reflected waves and perfect transmission ($T \rightarrow 1$), the thickness d_{match} of this layer must strictly equal one-quarter of the acoustic wavelength λ at the center frequency f_c of the multiplexed signal:

$$d_{match} = \frac{\lambda}{4} = \frac{v_{match}}{4f_c}$$

By doping a sub-layer of the h-BN epoxy to precisely match Z_{match} , the reflection coefficient R is driven to near zero, effectively neutralizing the exponential layer-by-layer signal attenuation.

2.8.2 Phononic Bandgap Engineering (Crosstalk Elimination)

To prevent lateral signal leakage (crosstalk) between adjacent Acoustic Through-Silicon Vias (A-TSVs), the silicon substrate surrounding each acoustic pillar is structured into a **Phononic Crystal (PnC)**. By etching a periodic lattice of microscopic air holes (or filling them with a low-density polymer) around the pillars, an acoustic bandgap is created.

Acoustic waves within this bandgap cannot propagate laterally through the lattice. The lattice constant a (distance between holes) is designed to satisfy the Bragg scattering condition for the transverse leakage waves:

$$a = \frac{v_{transverse}}{2f}$$

Waves attempting to cross into adjacent memory cells fall into this forbidden frequency band and are exponentially dampened out laterally, acting as an absolute acoustic isolator and drastically lowering the XT (Crosstalk) value.

3. Chemical Material System

- **Ru Top-Electrode:** 20 nm (Adhesion promoter and electrical contact)
 - **CoFe₂O₄:** 15 nm (Magnetostrictive layer, free of rare-earth elements)
 - **BaTiO₃:** 50 nm (Piezoelectric actuator, lead-free)
 - **SrRuO₃:** 30 nm (Bottom-Electrode / Buffer layer)
 - **AlN:** 80 nm (Acoustic transducer, MEMS-compatible)
 - **Si(100):** Highly doped silicon substrate
-

4. Fabrication Protocol (TRL 3 / TRL 4)

1. **Transducer:** Reactive magnetron sputtering of AlN at 350°C.
 2. **Bottom-Electrode:** Deposition of SrRuO₃ (Pulsed Laser Deposition).
 3. **Heterostructure:** Combined deposition of BaTiO₃ and CoFe₂O₄ under controlled oxygen partial pressure.
 4. **Patterning:** DUV lithography and ion beam etching (IBE).
 5. **Passivation & Microchannels:** ALD passivation and wet-chemical KOH etching on the wafer backside.
 6. **Coupling:** Spin-coating of the h-BN epoxy matrix followed by UV curing.
-

5. Thermodynamics and Microfluidics (FEM Basis)

The capillary pressure generated within the microchannel (Young-Laplace) is given by:

$$\Delta P_{cap} = \frac{2\gamma \cos(\theta_c)}{r_{eff}} \approx 10.71 \text{ kPa}$$

For future thermal Finite Element Method (FEM) analyses, the transient heat conduction equation is applied:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla(k \nabla T) + Q$$

Simulation Expectation: Target cooling capacity: $> 520 \text{ W/cm}^2$ (subject to future experimental validation). Validation of dry-out parameters is pending.

6. Economic & Yield Models

Cost projections are based on theoretical yield models assuming the elimination of extreme Cu-Cu hybrid bonding processes:

Poisson Model: $Y = e^{-DA}$

Murphy Model: $Y = \left(\frac{1-e^{-DA}}{DA}\right)^2$

Domain	HBM4 (Industry)	PCME-RAM (Modeling Assumption)
Wafer	128 USD	32 USD
Lithography	185 USD	38 USD
TSV/Bumps	94 USD	6.2 USD
Bonding	140 USD	12.5 USD
Cooling	45 USD	8.8 USD

7. Performance Comparison (Theoretical Projection)

Parameter	HBM4E (Industry Standard)	PCME-RAM (Research Goal)
Memory Type	DRAM	MeRAM
Refresh	Yes	No (Non-Volatile)
Cooling	Conventional	Microfluidics (Simulation Target)
Bandwidth	3.6–4.0 TB/s	Target Bandwidth 4.8 TB/s (Derived from the theoretical FDM capacity model)
Latency	~12 ns	<2.5 ns (Modeling Assumption)

8. Forensic Uncertainty and Risk Analysis

Component / Metric	Feasibility Status	Technical Risk / Comment
BaTiO3 / CoFe2O4 / AlN	Plausible & Available	Low (Materials are industrially established).
Microfluidics	Plausible	Medium (Dry-out effects remain unclear).
Acoustic Multiplexing	Experimental	High (Resonances/Attenuation losses).
TSV Substitution	Highly Risky	Critical (PDN compatibility is questionable).
4.8 TB/s & <2.5 ns	Unproven	Very High (Capacity model vs. reality).

9. Development Roadmap (Technology Readiness Levels)

- **TRL 3:** Single-cell verification via MOKE and PFM (Magnetic Statics).
- **TRL 4:** 8x8 array demonstrator and crosstalk measurements (Acoustic Impedance).
- **TRL 5:** 4-layer stack featuring microfluidics (Transient FEM validation).
- **TRL 6:** 16-layer engineering sample (Acquisition of empirical yield data).

10. Research Modeling and Feasibility Framework (OOA-SIM-V1)

Scientific Disclaimer regarding the Simulation:

The attached script OOA-SIM-V1 serves strictly as a visualization and feasibility framework for initial parameter studies. The implemented mathematical approximations (e.g., the sigmoid function representing LLG dynamics or the simplified exponential attenuation model) are intended for the conceptual demonstration of architectural targets. **They do not replace fully coupled, physical solvers such as Micromagnetics (OOMMF/MuMax3), Acoustic FDTD, or comprehensive thermal FEM simulations.**

```
# Lead Architect: Emanuel Schaaf
# Open Origin Architecture
# PCME-RAM Research Modeling and Feasibility Framework V1

import sys
import traceback

try:
    import numpy as np
    import matplotlib
    matplotlib.use("Agg")
    import matplotlib.pyplot as plt
    import matplotlib.animation as animation
except ImportError as e:
    print(f"Critical Import Error: {e}")
    input("Press Enter to safely exit the application...")
    sys.exit(1)

def run_simulation():
    # --- Style Configuration (Dark Mode) ---
    plt.style.use("dark_background")
    fig = plt.figure(figsize=(14, 12), facecolor="#0d1117")
    fig.suptitle("PCME-RAM: OOA-SIM-V1 Parameter Visualization",
    fontsize=16, color="#00ffcc", fontweight="bold")

    gs = fig.add_gridspec(3, 1, hspace=0.45)
    ax1 = fig.add_subplot(gs[0, 0], facecolor="#161b22")
    ax2 = fig.add_subplot(gs[1, 0], facecolor="#161b22")
    ax3 = fig.add_subplot(gs[2, 0], facecolor="#161b22")

    for ax in [ax1, ax2, ax3]:
        ax.grid(color="#30363d", linestyle="--", linewidth=0.5)
        ax.tick_params(colors="white")
        for spine in ax.spines.values():
            spine.set_color("#30363d")
```

```

# --- Data Generation (Framework Models) ---
frames = 150

# 1. LLG Dynamics Approximation (Sigmoid Reversal)
t_llg = np.linspace(0, 5, frames) # Time in ns
mz = -1 + 2 / (1 + np.exp(-3 * (t_llg - 2.5)))

# 2. BVD Equivalent Circuit (Impedance over Frequency)
freq = np.linspace(1e9, 10e9, frames) # 1 GHz to 10 GHz
omega = 2 * np.pi * freq
C0 = 2e-13
Cm = 1e-14
Lm = 1e-8
Rm = 5.0

Z_m = Rm + 1j * omega * Lm + 1 / (1j * omega * Cm)
Z_total = 1 / (1j * omega * C0 + 1 / Z_m)
Z_mag = np.abs(Z_total)

# 3. Acoustic Attenuation Model ( $A(x) = A_0 * e^{(-\alpha * x)}$ )
layers = np.linspace(1, 16, frames) # 1 to 16 Layers (Stack)
alpha_damping = 0.12 # Assumed attenuation coefficient
amplitude = 1.0 * np.exp(-alpha_damping * layers)

# --- Plot Setup ---
line1, = ax1.plot([], [], color="#00ffcc", lw=2, label="Mz
(Magnetization Reversal)")
ax1.set_xlim(0, 5)
ax1.set_ylim(-1.1, 1.1)
ax1.set_title("Modeled LLG Switching Dynamics (Magnetoelectric Strain)",
color="white")
ax1.set_ylabel("Normalized Mz", color="white")
ax1.set_xlabel("Time (ns)", color="white")
ax1.legend(loc="lower right", facecolor="#161b22", edgecolor="#30363d",
labelcolor="white")

line2, = ax2.plot([], [], color="#ff00ff", lw=2, label="|Z| Impedance
(Ohm)")
ax2.set_xlim(1, 10)
ax2.set_ylim(0, np.max(Z_mag) * 1.1)
ax2.set_title("BVD Equivalent Circuit (AlN Transducer Resonance)",
color="white")
ax2.set_ylabel("Magnitude |Z|", color="white")
ax2.set_xlabel("Frequency (GHz)", color="white")
ax2.legend(loc="upper right", facecolor="#161b22", edgecolor="#30363d",
labelcolor="white")

line3, = ax3.plot([], [], color="#ffd700", lw=2, label="Signal Amplitude
A(x)")
ax3.set_xlim(1, 16)

```

```

ax3.set_ylim(0, 1.1)
ax3.set_title("Acoustic Attenuation across Memory Layers ( $A(x) = A_0 e^{-\alpha x}$ )", color="white")
ax3.set_ylabel("Relative Amplitude", color="white")
ax3.set_xlabel("Stack Layer (x)", color="white")
ax3.legend(loc="upper right", facecolor="#161b22", edgecolor="#30363d",
labelcolor="white")

# --- Animation Function ---
def update(frame):
    # Progressive drawing process
    line1.set_data(t_llg[:frame], mz[:frame])
    line2.set_data(freq[:frame] / 1e9, Z_mag[:frame])
    line3.set_data(layers[:frame], amplitude[:frame])
    return line1, line2, line3

ani = animation.FuncAnimation(fig, update, frames=frames, interval=30,
blit=True)

# --- Export Logic ---
print("Initiating progressive animation export (High-Res MP4)...")
writer = animation.FFMpegWriter(fps=30, bitrate=3000)
try:
    ani.save("PCME_RAM_Simulation_V1.mp4", writer=writer)
    print("Success: Animation saved as 'PCME_RAM_Simulation_V1.mp4'")
except Exception as e:
    print(f"FFmpeg failed or is not installed: {e}")
    print("Failing over silently to PillowWriter (GIF)...")
    try:
        writer_gif = animation.PillowWriter(fps=30)
        ani.save("PCME_RAM_Simulation_V1.gif", writer=writer_gif)
        print("Success: Fallback animation saved as 'PCME_RAM_Simulation_V1.gif'")
    except Exception as e2:
        print(f"Pillow export also failed: {e2}")

plt.close(fig)

if __name__ == "__main__":
    print("Initializing Open Origin Architecture Framework...")
    try:
        run_simulation()
    except KeyboardInterrupt:
        print("\nProcess aborted by user (KeyboardInterrupt).")
    except Exception as e:
        print(f"\nCritical Runtime Error encountered:")
        traceback.print_exc()
    finally:

```

```
input("\nExecution completed. Press Enter to close the console window...")
```

Environment Setup & Execution Guide for OOA-SIM-V1

Lead Architect: Emanuel Schaaf

Project: PCME-RAM (Open Origin Architecture)

To ensure the `OOA-SIM-V1` simulation script executes flawlessly and generates the high-resolution progressive animations, your local Python environment must meet specific dependency requirements.

1. Python Package Dependencies (PIP)

The script relies on standard scientific libraries for mathematical array processing and headless plotting. Install the required packages via your terminal or command prompt:

Bash

```
pip install numpy scipy matplotlib pillow
```

Note: `pillow` is strictly required as a fallback rendering engine to guarantee the script can successfully output an animated `.gif` if the primary `MP4` encoder is unavailable.

2. System-Level Dependencies (FFmpeg)

To export the simulation as a high-resolution `.mp4` video (which is the default and preferred behavior of the script), the **FFmpeg** multimedia framework must be installed on your operating system and added to your system's `PATH`.

- **For Debian/Ubuntu Linux environments:** Bash

```
sudo apt update
sudo apt install ffmpeg
```

- **For Windows environments:** Download the pre-compiled FFmpeg binaries, extract them to your local drive (e.g., `C:\ffmpeg`), and add the `\bin` folder to your Windows Environment Variables (`Path`). Alternatively, install it via Winget: PowerShell

```
winget install ffmpeg
```

3. Execution Instructions

Because the script utilizes the `matplotlib.use("Agg")` directive, it runs in a completely headless state. It will not spawn any interactive GUI windows, making it safe to run on remote servers, Proxmox containers, or local headless nodes.

Execute the script via the terminal:

Bash

```
python pcme_sim_v1.py
```

Expected Output Sequence:

- 1. The script initializes the Open Origin Architecture Framework.
- 2. It processes the LLG, BVD, and Acoustic Attenuation arrays.
- 3. It hooks into FFmpeg to progressively render the 150 frames.
- 4. Upon success, it outputs `PCME_RAM_Simulation_V1.mp4` into the active directory.
- 5. A manual prompt (`Press Enter to close the console window...`) ensures the execution log remains visible for debugging purposes.

Category	Status
Architecture Definition	✓
Physical Models	✓
Material System	✓
Fabrication Concept	✓
Thermal Model	✓
Yield Model	✓
Uncertainty Model	✓
Risk Model	✓
Simulation Framework	✓
Scientific Integrity	✓
Experimental Proof	✗ (Pending external validation)
Industrial Validation	✗ (Pending external validation)

12. Conclusion

The PCME-RAM concept defines a highly innovative, hybrid memory architecture. It combines established physical principles (magnetoelectrics, piezoelectrics, elastodynamics) into a novel information technology construct. Through the integration of acoustic impedance and attenuation models, LLG dynamics, fundamental thermal FEM equations, realistic yield models, and the numerical modeling framework, this document operates as a comprehensive and validatable blueprint for the PCME research program within the Open Origin Architecture.

PCME-RAM Open-Source License

(PCME-RAM-OSL)

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